NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Technology

PROJECT PROPOSAL

KrishiBondhu AI – An Intelligent and Personalized Agricultural Decision Support System

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Submitted To

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1. Project Title

KrishiBondhu AI – An Intelligent and Personalized Agricultural Decision Support System

KrishiBondhu AI is a web-based agricultural assistance platform designed to provide accessible, AI-powered farming guidance to farmers in Bangladesh. The system will help users understand common agricultural problems and make better farming decisions through an interactive, Bangla-friendly interface.

2. Problem Statement

2.1 Existing Problem

Agriculture is an important part of Bangladesh's economy, and many people depend on farming for their livelihood. However, farmers frequently face difficulties in making appropriate farming decisions. They may lack access to reliable information about crop diseases, pests, fertilizers, soil conditions, irrigation, and suitable crops for different seasons.

Many farmers depend on traditional knowledge, local advice, or information from different sources. These sources may not always provide accurate, updated, or situation-specific guidance. Language barriers and limited access to agricultural experts can also make it difficult for farmers to identify and solve farming problems.

2.2 Importance of the Problem

Incorrect farming decisions can lead to crop damage, unnecessary expenses, reduced production, and financial losses. Easy access to understandable agricultural information can help farmers make more informed decisions, use resources efficiently, and improve their farming practices.

A simple digital platform that provides agricultural guidance in Bangla could make useful information more accessible to farmers, particularly those who have limited access to agricultural experts.

2.3 Limitations of Existing Solutions

Although agricultural information is available through websites, mobile applications, and government agricultural services, users may face several limitations:

- Information is often distributed across multiple platforms.

- Some platforms provide general information rather than guidance based on a farmer's specific situation.
- Farmers may find technical agricultural information difficult to understand.
- Identifying crop problems from symptoms or images may require expert assistance.
- Farmers may have difficulty keeping track of crop activities and previous recommendations.

Therefore, a simple and integrated agricultural assistance platform could help address these challenges.

2.4 Why AI Is Relevant

Artificial Intelligence can help users interact with agricultural information more easily. Instead of searching through multiple sources, farmers can describe their problems in natural language and receive understandable responses.

KrishiBondhu AI will use the Google Gemini API to provide AI-powered agricultural question answering and preliminary crop image analysis. The system will also use structured user inputs, such as crop type, location, season, and farming conditions, to provide more relevant recommendations.

The project will use an existing AI API rather than training a machine learning model from scratch. This approach will reduce development complexity, cost, and time.

3. Project Objectives

The main objective of KrishiBondhu AI is to develop an accessible AI-powered agricultural assistance platform for farmers in Bangladesh.

The specific objectives are:

1. To provide agricultural information and guidance through a Bangla-friendly AI chatbot.
2. To help users understand common crop diseases and problems through image-based preliminary analysis.
3. To provide basic crop and fertilizer recommendations based on user-provided information.
4. To provide information about common agricultural pests and their possible management methods.
5. To help farmers maintain basic digital records of their farming activities.
6. To make agricultural information easier to access through a simple and user-friendly web interface.

7. To encourage informed farming decisions and reduce unnecessary use of agricultural resources.

4. Proposed Features

The proposed system will include the following features:

4.1 User Registration and Login

Users will be able to create accounts and log in to the platform. Their basic information, farming records, and saved recommendations will be stored securely.

4.2 AI Agricultural Assistant

An interactive AI chatbot will allow users to ask agricultural questions in Bangla or English.

Users will be able to ask questions about:

- Crop cultivation and care
- Soil preparation and irrigation
- Fertilizer usage
- Common crop diseases
- Pest identification and management
- General farming practices

The chatbot will use the Google Gemini API to generate understandable responses.

4.3 Crop Disease Image Analysis

Users will be able to upload photographs of affected crops or leaves. The AI integration will analyze the uploaded image and provide preliminary information about possible crop diseases or visible symptoms.

The system will also provide basic suggestions for further inspection and possible next steps.

This feature will be presented as preliminary AI-assisted analysis, not a guaranteed diagnosis.

4.4 Fertilizer Recommendation

Users will provide information such as crop type, growth stage, and available soil information. Based on the provided information and available agricultural guidance, the system will generate basic fertilizer-related suggestions.

The system will avoid presenting unverified fertilizer quantities as scientifically confirmed recommendations.

4.5 Crop Recommendation

The platform will provide basic crop suggestions based on information such as:

- Location
- Season
- Soil type
- Available farming conditions

This feature will help users explore crops that may be suitable for their farming situation.

4.6 Pest Management Guidance

Users will be able to ask questions about common agricultural pests and receive information about their identification, possible effects, and general management practices.

4.7 Farm Record Management

Users will be able to maintain basic digital records of their farming activities, including:

- Crop name
- Planting date
- Farming notes
- Important observations

This feature will help farmers organize their agricultural information.

4.8 Weather Information

The platform may provide weather information through an external weather API. This information can help users consider weather conditions when planning irrigation and other farming activities.

4.9 Saved Recommendations and Chat History

Users will be able to access previous conversations and save useful AI-generated recommendations for future reference.

4.10 User Dashboard

A simple dashboard will provide access to the main features, including the AI assistant, crop image analysis, farm records, and saved information.

The project will use the following technologies:

5. Technology Stack

KrishiBondhu AI will be developed using modern web development technologies and an existing Artificial Intelligence API. The selected technologies will help the team build a responsive, interactive, and cost-effective agricultural assistance platform. The technology stack will cover frontend development, backend operations, database management, AI integration, version control, and deployment.

5.1 Technologies Overview

The following technologies will be used to develop the proposed system:

Component	Technology	Purpose
Frontend	React.js	Developing the user interface
Styling	CSS / Tailwind CSS	Designing a responsive and user-friendly interface
Backend Runtime	Node.js	Handling server-side operations
Backend Framework	Express.js	Developing REST APIs
Database	MongoDB Atlas	Storing user information and farming records
AI Integration	Google Gemini API	Providing AI-powered agricultural assistance
Version Control	Git and GitHub	Managing source code and team collaboration
Frontend Deployment	Vercel	Hosting the frontend application
Backend Deployment	Render	Hosting the backend application

Technology Stack table

5.2 Frontend Technologies

React.js

React.js will be used to develop the frontend of KrishiBondhu AI. It will help create an interactive and responsive user interface. The application will include different pages and components for the AI assistant, crop image analysis, farm records, and user dashboard.

CSS / Tailwind CSS

CSS and Tailwind CSS will be used to design the application's layout, colors, typography, and overall appearance. These technologies will help create a simple, clean, and mobile-friendly interface that users can access from different devices.

5.3 Backend Technologies

Node.js

Node.js will be used as the backend runtime environment. It will handle server-side operations and manage communication between the frontend, database, and external AI services.

Express.js

Express.js will be used as the backend framework for Node.js. It will help develop REST APIs for user authentication, data management, AI requests, and communication between different parts of the application.

5.4 Database Technology

MongoDB Atlas

MongoDB Atlas will be used as the cloud-based database solution for KrishiBondhu AI. It will store and manage the application's required information.

The database may contain:

- User account information.
- Crop and farming records.
- Saved AI recommendations.
- Previous conversations and chat history.
- Other relevant application data.

MongoDB Atlas will provide a flexible database structure that can support the application's development requirements.

5.5 Artificial Intelligence Integration

Google Gemini API

The Google Gemini API will be integrated into KrishiBondhu AI to provide AI-powered agricultural assistance. It will help users interact with the system using natural language and receive understandable responses in Bangla or English.

The AI integration will support the following functionalities:

- Answering agricultural questions related to crops, soil, fertilizers, and irrigation.
- Providing preliminary analysis of uploaded crop images.
- Generating basic crop-related suggestions based on user-provided information.
- Providing general guidance about common crop diseases and pest management.

The project will use the existing Google Gemini API instead of training a custom AI model. This approach will reduce development complexity, cost, and implementation time.

AI-generated recommendations will be treated as preliminary guidance. Users will be encouraged to consult agricultural experts for important farming decisions.

5.6 Version Control and Deployment

Git and GitHub

Git will be used for version control to track changes in the project's source code. GitHub will be used as the central repository for storing the project and supporting collaboration among team members.

All team members will contribute to the repository, maintain their development work, and regularly update the project code.

Vercel

Vercel will be used for deploying and hosting the frontend application. It will make the web application accessible to users through an online URL.

Render

Render will be used for deploying and hosting the backend server developed with Node.js and Express.js. It will handle backend requests and communication with the database and AI services.

The team will consider available free or low-cost hosting plans to keep deployment expenses within the project budget.

6. Expected Outcome

At the end of the project, the team expects to develop a functional web-based agricultural assistance platform that provides AI-powered farming guidance.

The expected outcomes include:

6.1 Benefits to Users

- Farmers will be able to access agricultural information through a simple web application.
- Users will be able to ask farming-related questions in Bangla or English.
- Farmers will receive preliminary information about possible crop diseases through image analysis.
- Users will be able to explore basic crop and fertilizer recommendations.
- Farmers will be able to maintain digital records of their farming activities.
- Useful recommendations and previous conversations will be easier to access.

6.2 Workflow Improvement

The platform will bring several agricultural assistance features together in one place. Users will not need to depend entirely on searching through different websites or manually organizing their farming information.

The AI assistant will make information retrieval more interactive, while the farm record feature will help users maintain organized agricultural records.

6.3 Benefits of AI Integration

AI integration will allow the platform to understand natural-language questions, generate context-based responses, and analyze uploaded crop images.

It will make agricultural information more interactive and accessible. However, AI-generated information may not always be accurate. Users will be advised to consult qualified agricultural experts or local agricultural extension officers before making important farming decisions.

7. Initial Project Planning

7.1 Development Approach

The project will follow an incremental development approach. The team will develop the core features first, test them, and gradually integrate additional features.

The development process will include requirement analysis, interface design, backend development, AI integration, testing, and deployment.

7.2 Development Phases

Phase	Activities
Phase 1: Requirement Analysis	Identify user requirements and finalize the project scope
Phase 2: UI Design	Design the website layout and user interface
Phase 3: Frontend Development	Develop the main pages and user dashboard using React.js
Phase 4: Backend Development	Develop REST APIs, authentication, and database operations
Phase 5: AI Integration	Integrate the Gemini API for agricultural chat and image analysis
Phase 6: Testing and Integration	Test individual features and integrate the complete system
Phase 7: Deployment	Deploy the application and prepare project documentation

Development Phases table

7.3 Team Responsibilities

The team will divide responsibilities according to members' skills and project requirements.

Role	Responsibility
Frontend Developer	Project coordination, React.js development, interface design, and integration
Team Leader & Backend & Database Developer & AI Integration	Node.js, Express.js, REST APIs, and MongoDB integration, Gemini API integration
Testing Lead	Prompt development, testing, and documentation

Team Responsibilities table

7.4 Initial Development Scope

Considering the team's current development experience and available time, the project will prioritize the following core features:

1. User registration and login.
2. AI-powered agricultural chatbot.
3. Crop image analysis.
4. Basic crop and fertilizer recommendations.
5. Farm record management.
6. Responsive user dashboard.

Additional features, such as weather information, pest management guidance, reminders, and advanced history management, will be implemented if sufficient time and resources are available.

7.5 Potential Challenges and Risk Management

Potential Challenge	Proposed Solution
Inaccurate AI-generated information	Use carefully designed prompts and reliable agricultural references; advise expert consultation when necessary
Poor-quality crop images	Provide image upload guidelines and explain the limitations of image analysis
Limited API usage	Monitor API usage and use available free-tier resources efficiently
Limited development experience	Divide tasks into smaller modules and follow incremental development

Deployment limitations	Use free or low-cost hosting services where available

Risk Management table

The team will prioritize a functional and reliable core application rather than attempting to implement too many features within the available development period.

8. Conclusion

KrishiBondhu AI aims to make agricultural information more accessible and understandable for farmers in Bangladesh through an AI-powered web application. By combining an agricultural chatbot, preliminary crop image analysis, basic farming recommendations, and digital farm records, the platform will provide several useful agricultural assistance services in one place.

The project will use existing AI services and commonly used web development technologies to keep implementation practical and affordable. The team expects to develop a functional prototype that demonstrates how AI can support agricultural decision-making and contribute to addressing real-world farming challenges.

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